Numerical Solutions to Partial Differential Equations

《偏微分方程数值解》

n-D 1st Order Linear Hyperbolic Partial Differential Equation

1. Scalar case ($u \in \mathbb{R}^1$), $$u_t + \sum_{i=1}^{n} a_i\, u_{x_i} + b\, u = \psi_0, \qquad (3.1.1)$$
where a_i, b, ψ_0 are real functions of $x = (x_1, \ldots, x_n)$ and t.
2. Vector case ($\mathbf{u} = (u_1, \cdots, u_p)^T \in \mathbb{R}^p$),
$$\mathbf{u}_t + \sum_{i=1}^{n} A_i\, \mathbf{u}_{x_i} + B\, \mathbf{u} = \boldsymbol{\psi}_0, \qquad (3.1.2)$$
where A_i, $B \in \mathbb{R}^{p\times p}$, $\boldsymbol{\psi}_0 \in \mathbb{R}^p$ are real functions of t and $x = (x_1, \ldots, x_n)$, and $\forall \boldsymbol{\alpha} \in \mathbb{R}^n$, $A(x,t) = \sum_{i=1}^n \alpha_i A_i(x,t)$ is real diagonalizable, *i.e.* $A(x,t)$ has p linearly independent eigenvectors corresponding to real eigenvalues.
3. If $A(x,t) = \sum_{i=1}^n \alpha_i A_i(x,t)$ has p mutually different real eigenvalues, the system is called strictly hyperbolic.

p个相互不同的实特征值

n-D 2nd Order Linear Hyperbolic Partial Differential Equation

- A general 2nd order scalar equation ($u \in \mathbb{R}^1$),

$$u_{tt} + 2\sum_{i=1}^{n} a_i\, u_{x_i t} + b_0\, u_t - \sum_{i,j=1}^{n} a_{ij}\, u_{x_i x_j} + \sum_{i=1}^{n} b_i\, u_{x_i} + cu = \psi_0, \tag{3.1.3}$$

  where a_i, a_{ij}, b_i, c and ψ_0 are real functions $x = (x_1, \ldots, x_n)$ and t, (a_{ij}) is real symmetric positive definite.

- Define $v = u$, $v_0 = u_t$, $v_i = u_{x_i}$, then the above 2nd order scalar equation transforms into a first order linear system of partial differential equations for $\mathbf{v} = (v, v_0, v_1, \cdots, v_n)^T$

$$A\mathbf{v}_t + \sum_{i=1}^{n} A_i \mathbf{v}_{x_i} + B\mathbf{v} = \psi_0, \tag{3.1.4}$$

n-D 2nd Order Scalar Transforms to n-D 1st Order System ($p = n + 2$)

$$A = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & a_{n1} & \cdots & a_{nn} \end{bmatrix}, \qquad A_i = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 2a_i & -a_{1i} & \cdots & -a_{ni} \\ 0 & -a_{1i} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & -a_{ni} & & & \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & -1 & 0 & \cdots & 0 \\ c & b_0 & b_1 & \cdots & b_n \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}, \qquad \boldsymbol{\psi}_0 = \begin{bmatrix} 0 \\ \psi_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

n-D 2nd Order Scalar Transforms to n-D 1st Order System ($p = n + 2$)

1. Let $R^T AR = I$ (A is real symmetric positive definite);
2. Introduce a new variable $\mathbf{w} = R^{-1}\mathbf{v}$;
3. Denote $\hat{A}_i = R^T A_i R$, $\hat{B} = R^T BR$, $\hat{\psi}_0 = R^T \psi$;
4. $\hat{A}(x, t) = \sum_{i=1}^n \alpha_i \hat{A}_i(x, t)$ is a real symmetric matrix and thus is real diagonizable for all real α_i, $i = 1, \ldots, n$;
5. The 2nd order scalar equation now transforms into a 1st order linear hyperbolic system of partial differential equations for $\mathbf{w} \in \mathbb{R}^{(n+2)}$:
$$\mathbf{w}_t + \sum_{i=1}^{n} \hat{A}_i \mathbf{w}_{x_i} + \hat{B}\mathbf{w} = \hat{\psi}_0. \qquad (3.1.5)$$

Standard Form of *n*-D 1st Order Linear Hyperbolic Equations

1. The standard form of 1st order linear hyperbolic equation:

$$u_t + \sum_{i=1}^{n} a_i \, u_{x_i} = \psi, \quad (\psi = \psi_0 - b\,u). \qquad (3.1.6)$$

2. The standard form of 1st order linear hyperbolic system:

$$\mathbf{u}_t + \sum_{i=1}^{n} A_i \, \mathbf{u}_{x_i} = \boldsymbol{\psi}, \quad (\boldsymbol{\psi} = \boldsymbol{\psi}_0 - B\,\mathbf{u}); \qquad (3.1.7)$$

3. The equation (system) is said to be homogeneous, if $\psi = 0$;

Standard Form of *n*-D 1st Order Linear Hyperbolic Equations

4. In general, a higher order linear hyperbolic equation (system of equations) can always be transformed into a first order linear hyperbolic system of equations.

5. An equation (system) is said to be nonlinear, if at least one of the coefficients depends on the unknown or the right hand side term is a nonlinear function of the unknown.

An Example of a Balance Law

Substance balance in flowing fluid in a 1D pipe.

1. $u(x,t)$: substance density, measured by mass per unit length.
2. $f(x,t,u)$: mass flux, measured by mass per unit time.
3. $\psi(x,t,u)$: the density of the mass source (or sink), measured by mass per unit length per unit time.
4. Balance law (integral form), for given $x_l < x_r$ and $t_b < t_a$,

$$\int_{x_l}^{x_r} u(x,t_a)\,dx = \int_{x_l}^{x_r} u(x,t_b)\,dx + \int_{t_b}^{t_a} f(x_l,t,u(x_l,t))\,dt$$
$$- \int_{t_b}^{t_a} f(x_r,t,u(x_r,t))\,dt + \int_{t_b}^{t_a}\int_{x_l}^{x_r} \psi(x,t,u(x,t))\,dx\,dt.$$

An Example of a Balance Law

5. Suppose u and f are sufficiently smooth, we are led to

$$\int_{t_b}^{t_a}\int_{x_l}^{x_r} [u(x,t)_t + f(x,t,u(x,t))_x - \psi(x,t,u(x,t))]\, dx\, dt = 0,$$

for all given $x_l < x_r$ and $t_b < t_a$;

6. or equivalently:

$$u(x,t)_t + f(x,t,u(x,t))_x = \psi(x,t,u(x,t)),$$

this is the balance law in differential form.

Advection Equation and Advection-Diffusion Equation

The simplest example is the 1D advection equation:

$$u_t + a\, u_x = 0,$$

in which the flowing velocity of the fluid is a constant a, the flux is simply $f(x, t, u) = f(u) = a\, u$, and the source term is zero.

Another simple example is the 1D advection-diffusion equation:

$$u_t + a\, u_x = \sigma\, u_{xx},$$

in which diffusion as well as advection is considered, and the flux is of the form $f(x, t, u, u_x) = a\, u - \sigma\, u_x$, where σ is the diffusion parameter.

Note: Advection equation is also called convection equation, advection-diffusion equation is also called convection-diffusion equation.

Characteristics and Riemann Invariants

In hyperbolic equations (systems), the information propagates at finite characteristic speeds.

Consider a 1D constant-coefficient hyperbolic system

$$\mathbf{u}_t + A\,\mathbf{u}_x = 0.$$ homogeneous

1. $AR = R\Lambda$, $\Lambda = \mathrm{diag}(\lambda_1, \cdots, \lambda_p)$ with $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_p$.
2. Define $\mathbf{w} = R^{-1}\mathbf{u}$, the system is transformed to
$$\mathbf{w}_t + \Lambda\,\mathbf{w}_x = 0.$$
3. p families of characteristics given by p characteristic equations
$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \lambda_i, \qquad i = 1, \cdots, p.$$

Riemann Invariants and General Solutions of Initial Value Problems

4. w_i is a constant on each characteristics of the ith family:

$$\frac{\mathrm{d}w_i(x_i(t),t)}{\mathrm{d}t} = [w_{i,t} + \lambda_i\, w_{i,x}]\,(x_i(t),t) = 0.$$

In general, it holds for
the homogeneous pde.

5. $w_i(x,t) = w_i(x - \lambda_i t, 0)$, $i = 1, \cdots, p$.

characteristic variables

6. w_i, $i = 1, \cdots, p$, are called the Riemann invariants of the system with respect to the characteristics of (λ_i, ξ^i), with $A\xi^i = \lambda_i \xi^i$, $i = 1, \cdots, p$.

Riemann Invariants and General Solutions of Initial Value Problems

7. Denote $\boldsymbol{\xi}^i$ the ith column of R, and $\boldsymbol{\eta}^i$ the ith row of R^{-1}.

8. By definition $\boldsymbol{\eta}^i \mathbf{u} = w_i$, in particular $w_i(x,0) = \boldsymbol{\eta}^i \mathbf{u}^0(x)$.

9. By definition $\mathbf{u}(x,t) = R\mathbf{w}(x,t)$, thus, by ⑤ and ⑧,

$$\mathbf{u}(x,t) = \sum_{i=1}^{p} \boldsymbol{\xi}^i w_i(x,t) = \sum_{i=1}^{p} \boldsymbol{\xi}^i \boldsymbol{\eta}^i \mathbf{u}^0(x - \lambda_i t).$$

Domains of Dependance, Influence and Determination

Since $\mathbf{u}(x,t) = \sum_{i=1}^{p} \boldsymbol{\xi}^i \boldsymbol{\eta}^i \mathbf{u}^0(x - \lambda_i t)$, we define

1. domain of dependance of a set $\Omega_T \subset \mathbb{R} \times \mathbb{R}_+$:
   $D(\Omega_T) = \{y \in \mathbb{R} : y = x - \lambda_i t, \ \ i = 1, \cdots, p, \ \ \forall (x,t) \in \Omega_T\}$.
2. domain of influence of a set $\Omega \subset \mathbb{R}$:
   $I(\Omega) = \{(x,t) \in \mathbb{R} \times \mathbb{R}_+ : \ \exists\, 1 \le i \le p, \ s.t.\ x - \lambda_i t \in \Omega\}$,
3. domain of determination of a set $\Omega \subset \mathbb{R}$:
   $K(\Omega) = \{(x,t) \in \mathbb{R} \times \mathbb{R}_+ : x_i = x - \lambda_i t \in \Omega, \ \ i = 1, \cdots, p\}$.
4. We can also define $D(\Omega_T, t_0)$, $I(\Omega, t_0)$ and $K(\Omega, t_0)$ by replacing $\lambda_i t$ by $\lambda_i(t - t_0)$ in the above definitions.

Boundary Conditions in Initial-Boundary Value Problems

For an initial-boundary value problem defined on $(0,1) \times \mathbb{R}_+$, suppose $\lambda_1 < \ldots < \lambda_l < 0 < \lambda_r < \lambda_p$ ($l = r - 1$ or $l = r - 2$).

By the characteristics of the system, boundary conditions should be imposed in the following way:

1. $p - r + 1$ linearly independent boundary conditions on the left boundary 0, since $\{w_i\}_{i=r}^{p}$ picks up information from left.
2. l linearly independent boundary conditions on the right boundary 1, since $\{w_i\}_{i=1}^{l}$ picks up information from right.

The Characteristic Method for the Advection Equation

1. Advection equation: $u_t(x,t) + a(x,t)\, u_x(x,t) = 0$, $x \in I = (x_l, x_r) \subset \mathbb{R}$, $t > 0$;
2. Characteristics: $x'(t) = a(x,t)$, $x \in I$, $t > 0$;
3. Solution is a constant on a characteristic curve:
$$\frac{\mathrm{d}u(x(t),t)}{\mathrm{d}t} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\partial u}{\partial t} + a\,\frac{\partial u}{\partial x} = 0.$$
4. Suppose $a(x_l,t) > 0$ and $a(x_r,t) > 0$ for all $t \geq 0$.

The Characteristic Method for the Advection Equation

In particular, if $a(x,t) \equiv a$ is a constant, then, we have

9. for $a > 0$,

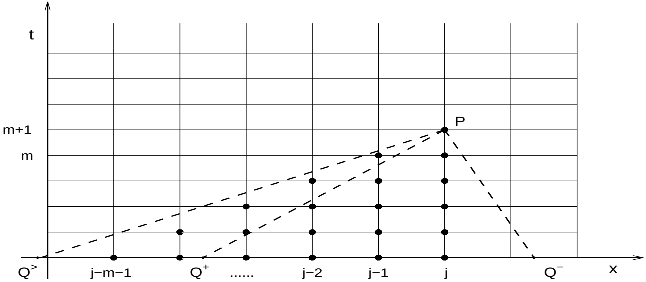
$$u(x,t) = \begin{cases} u^0(x - at), & x - at \geq x_l, \\ u_l\left(t - \dfrac{x - x_l}{a}\right), & x - at < x_l. \end{cases}$$

10. if $a < 0$,

$$u(x,t) = \begin{cases} u^0(x - at), & x - at \leq x_r, \\ u_r\left(t - \dfrac{x - x_r}{a}\right), & x - at > x_r. \end{cases}$$

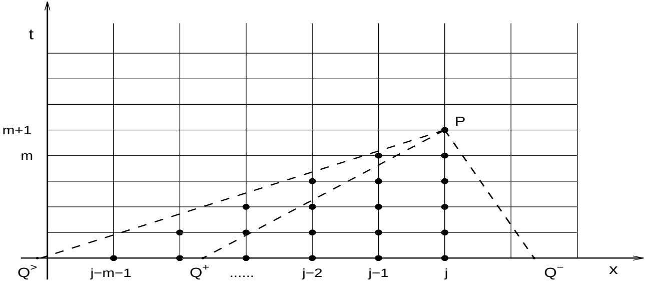
Domain of Dependance of a Difference Scheme

1. Advection equation: $u_t(x,t) + a\,u_x(x,t) = 0$, $(a = \text{constant})$.
2. The simplest scheme: $\dfrac{U_j^{m+1} - U_j^m}{\tau} + a\,\dfrac{U_j^m - U_{j-1}^m}{h} = 0;$
3. or equivalently: $U_j^{m+1} = (1-\nu)U_j^m + \nu U_{j-1}^m$, $(\nu = a\tau/h)$.
4. For $P = (x_j, t_{m+1})$, $D(P) = Q = x_j - a\,t_{m+1}$, the domain of dependance of the scheme $D_h(P) = \{x_{j-m-1}, \cdots, x_{j-1}, x_j\}$.



The CFL Condition of a Difference Scheme

- Simple observation: if $Q = Q^{>} < x_{j-m-1}$, or $Q = Q^{-} > x_j$, then U_j^{m+1} can not properly approximate $u(x_j, t_{m+1})$.
- **CFL condition**: a necessary condition for a numerical scheme to converge is that the domain of dependence of the the real solution is contained, at least in the sense of limit, in the domains of dependence of the numerical scheme.



The CFL Condition is a Necessary Condition for Stability

The CFL condition is often used as a necessary condition for the stability of finite difference schemes for hyperbolic differential equations.

In general, the CFL condition is not a sufficient condition for the stability of a numerical method.

CFL Condition is not a Sufficient Condition for Stability

Consider the finite difference scheme of the advection equation

$$\frac{U_j^{m+1} - U_j^m}{\tau} + a\,\frac{U_{j+1}^m - U_{j-1}^m}{2h} = 0, \quad a \in R^1 \setminus \{0\}\ .$$

The CFL condition for the scheme is $|\nu| = |a|\tau/h \leq 1.$

The Fourier modes solution $U_j^m = \lambda_k^m e^{\mathrm{i}\frac{kj\pi}{N}}$ with amplification factor $\lambda_k = 1 - \mathrm{i}\,\nu \sin\frac{k\pi}{N}$. Since $|\lambda_k| > 1$ for all $\nu \neq 0$, the scheme is always unstable, whether the CFL condition is satisfied or not.

Note: For initial value problems or initial-boundary value problems with periodic boundary conditions of a constant coefficient advection equation, a necessary and sufficient condition for L^2-stability is the so called von Neumann condition: $|\lambda_k| \leq 1 + K\tau$, for all $-N+1 \leq k \leq N$.

The Upwind Scheme for the Advection Equation

For the advection equation $u_t(x,t) + a(x,t)\,u_x(x,t) = 0$, by the CFL condition, the simplest difference scheme is

$$U_j^{m+1} = \begin{cases} U_j^m - \nu_j^m \triangle_+ U_j^m, & \text{if} \ \ a_j^m \le 0, \\ U_j^m - \nu_j^m \triangle_- U_j^m, & \text{if} \ \ a_j^m \ge 0, \end{cases}$$

where $\nu_j^m = a_j^m \,\tau / h$ satisfies $|\nu_j^m| \le 1$, or equivalently

$$U_j^{m+1} = \begin{cases} (1+\nu_j^m)U_j^m - \nu_j^m U_{j+1}^m, & \text{if} \ \ a_j^m \le 0, \\ (1-\nu_j^m)U_j^m + \nu_j^m U_{j-1}^m, & \text{if} \ \ a_j^m \ge 0, \end{cases}$$

which may be viewed as obtained by using

- $\triangle_{+t}/\triangle t$ to approximate $\partial/\partial t$;
- $\triangle_{+x}/\triangle x$ $(a < 0)$ or $\triangle_{-x}/\triangle x$ $(a > 0)$ to approximate $\partial/\partial x$.

The Upwind Scheme for the Advection Equation

Characteristic method + Linear interpolation $\Rightarrow$ Upwind scheme.

Let $a > 0$ be a constant, assume $\nu = a\tau/h \leq 1$.

1. By the characteristic method: $u(x_j, t_{m+1}) = u(x_j - a\,\tau, t_m)$;
2. By linear interpolation:
$$u(x_j - a\,\tau, t_m) \approx \frac{x_j - (x_j - a\tau)}{h} u_{j-1}^m + \frac{(x_j - a\tau) - x_{j-1}}{h} u_j^m;$$
3. Hence $u(x_j, t_{m+1}) \approx \nu u_{j-1}^m + (1-\nu) u_j^m$;
4. This leads to the scheme: $U_j^{m+1} = (1-\nu)U_j^m + \nu U_{j-1}^m$, which is exactly the upwind scheme.

The Truncation Error of the Upwind Scheme $O(\tau + h)$

By Taylor series expansion, we have

$$
\begin{aligned}
T_j^m := & \frac{u_j^{m+1} - u_j^m}{\tau} + a\,\frac{u_j^m - u_{j-1}^m}{h} - \left[\frac{\partial u}{\partial t} + a\,\frac{\partial u}{\partial x}\right]_j^m \\
= & \frac{1}{2}\left[\tau\, u_{tt} - a\, h\, u_{xx}\right]_j^m + \frac{1}{6}\left[\tau^2 u_{ttt} + a\, h^2\, u_{xxx}\right]_j^m + \cdots \\
= & -\left[\frac{1}{2} a\, h\,(1-\nu) u_{xx} + \frac{1}{6} a\, h^2 (1-\nu^2) u_{xxx} + \cdots\right]_j^m, \text{ if } a > 0.
\end{aligned}
$$

Note, here a is a constant, so $u_{tt} = a^2 u_{xx}$ and $\nu = a\tau/h$, and $\tau u_{tt} = ah\nu u_{xx}$. Similarly $\tau^k \partial_t^{k+1} u = (-1)^{k+1} a h^k \nu^k \partial_x^{k+1} u$.

The Truncation Error of the Upwind Scheme $O(\tau + h)$

Similarly, we have

$$\begin{aligned} T_j^m := & \frac{u_j^{m+1} - u_j^m}{\tau} + a\,\frac{u_{j+1}^m - u_j^m}{h} - \left[\frac{\partial u}{\partial t} + a\,\frac{\partial u}{\partial x}\right]_j^m \\ = & \frac{1}{2}\left[\tau\, u_{tt} + a\,h\,u_{xx}\right]_j^m + \frac{1}{6}\left[\tau^2 u_{ttt} + a\,h^2\,u_{xxx}\right]_j^m + \cdots \\ = & \left[\frac{1}{2}a\,h\,(1+\nu)u_{xx} + \frac{1}{6}a\,h^2(1-\nu^2)u_{xxx} + \cdots\right]_j^m, \text{ if } a < 0. \end{aligned}$$

Note, here a is a constant, so $u_{tt} = a^2 u_{xx}$ and $\nu = a\tau/h$, and $\tau u_{tt} = ah\nu u_{xx}$. Similarly $\tau^k \partial_t^{k+1} u = (-1)^{k+1} a h^k \nu^k \partial_x^{k+1} u$.

The Stability and Convergence of the Upwind Scheme

The error $e_j^m = U_j^m - u_j^m$ satisfies the following error equation

$$e_j^{m+1} = \begin{cases} (1-|\nu|)e_j^m + |\nu| e_{j+1}^m - \tau T_j^m, & \text{if } \ a < 0, \\ (1-|\nu|)e_j^m + |\nu| e_{j-1}^m - \tau T_j^m, & \text{if } \ a > 0. \end{cases}$$

If the CFL condition is satisfied, *i.e.* $|\nu| \le 1$, then, we have

$$E^{m+1} := \max_j |e_j^{m+1}| \le E^m + \tau \max_j |T_j^m| \le E^0 + t_{\max} \max_{m,j} |T_j^m|,$$
$$\forall (m+1)\tau \le t_{\max}.$$

The Solutions to Nonlinear Hyperbolic Equations are Generally not Smooth

Therefore, if the CFL condition is satisfied, the upwind scheme is stable in $\mathbb{L}^\infty$ norm, and its convergence rate is $O(\tau + h)$, **if** the solution u is sufficiently smooth.

However, in hyperbolic problems, discontinuities are commonly seen in the physical solutions. In fact, the discontinuity creation and propagation is a focus of investigation on nonlinear problems.

Nonlinear Advection Equation and Blow Up of Its Classical Solution

1. Nonlinear advection equation:
   $u_t(x,t) + a(u(x,t))\, u_x(x,t) = 0;$
2. Characteristic equation: $x'(t) = a(u(x,t)),\ t > 0;$
3. Solution is a constant on a characteristic curve:

$$\frac{\mathrm{d}u(x(t),t)}{\mathrm{d}t} = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\partial u}{\partial t} + a\,\frac{\partial u}{\partial x} = 0.$$

4. On a characteristic curve, $a(u(x,t))$ is a constant.
5. Characteristic curves are straight lines (with different slopes).
6. Suppose for $\xi_1 < \xi_2$, $a_1 := a(u^0(\xi_1)) > a_2 := a(u^0(\xi_2))$, then, at time $\bar{t} = (\xi_2 - \xi_1)/(a_1 - a_2) > 0$, the two corresponding characteristic lines intersect: $(\xi_1 + a_1\bar{t}, \bar{t}) = (\xi_2 + a_2\bar{t}, \bar{t})$.
7. In such a case, the classical solution blows up.

Weak Solutions for Nonlinear Conservation Laws

Since discontinuities are inevitable, we should take it into consideration by introducing the concept of weak solutions.

Definition

$u \in \mathbb{L}^1_{loc}(\mathbb{R} \times \mathbb{R}_+)$ is called a weak solution to the following 1D initial value problem of the conservation law

$$\begin{aligned} u_t(x,t) + f(u(x,t))_x &= 0, \\ u(x,0) &= u^0(x), \end{aligned}$$

if u satisfies the initial condition, and for all fixed $0 \le t_b < t_a$ and $-\infty < x_l < x_r < \infty$, the following equation holds

$$\int_{x_l}^{x_r} u(x,t_a)dx = \int_{x_l}^{x_r} u(x,t_b)dx + \int_{t_b}^{t_a} f(u(x_l,t))dt - \int_{t_b}^{t_a} f(u(x_r,t))dt.$$

Shock Speed and Rankine-Hugoniot Jump Condition

1. Suppose there is an isolated discontinuity (*i.e.* a shock) $x(t)$ propagating at speed $s(t)$ (*i.e.* $x'(t) = s(t)$);
2. By the integral form conservation law, for $x_1 = x(t_1)$ and $x_1 + \triangle x = x(t_1 + \triangle t)$, we have

$$\int_{x_1}^{x_1+\triangle x} u(x, t_1 + \triangle t)dx - \int_{x_1}^{x_1+\triangle x} u(x, t_1)dx$$
$$= \int_{t_1}^{t_1+\triangle t} f(u(x_1, t))dt - \int_{t_1}^{t_1+\triangle t} f(u(x_1 + \triangle x, t))dt.$$

3. Denote $u_l = u(x_1 - 0, t_1)$, $u_r = u(x_1 + 0, t_1)$, then, we have

$$(u_l - u_r)\triangle x = (f(u_l) - f(u_r))\triangle t + O(\triangle t^2).$$

4. Since $x'(t) = s(t)$, we are led to the Rankine-Hugoniot jump condition: $s[u] = [f]$.

Weak Solutions of Non-viscous Burger's Equation

Consider the initial value problem of the Burgers equation

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} = 0; \qquad u(x,0) = \begin{cases} 1, & x < 0; \\ 0, & x \geq 0. \end{cases}$$

- By the Rankine-Hugoniot jump condition,

$$s = \frac{f(u_r) - f(u_l)}{u_r - u_l} = \frac{\frac{1}{2}u_r^2 - \frac{1}{2}u_l^2}{u_r - u_l} = \frac{0 - \frac{1}{2}}{0 - 1} = \frac{1}{2}.$$

- Hence the weak solution of the problem is given by

$$u(x,t) = \begin{cases} 1, & x < t/2; \\ 0, & x \geq t/2. \end{cases}$$

Weak Solutions of Non-viscous Burger's Equation

Consider the initial value problem of the equation

$$\frac{\partial u^2}{\partial t} + \frac{2}{3}\frac{\partial u^3}{\partial x} = 0; \qquad u(x,0) = \begin{cases} 1, & x < 0; \\ 0, & x \geq 0. \end{cases}$$

- Define $w = u^2$, by the Rankine-Hugoniot jump condition,

$$s = \frac{f(w_r) - f(w_l)}{w_r - w_l} = \frac{\frac{2}{3}w_r^{3/2} - \frac{2}{3}w_l^{3/2}}{w_r - w_l} = \frac{0 - \frac{2}{3}}{0 - 1} = \frac{2}{3}.$$

- Hence the weak solution of the problem is given by

$$u(x,t) = \begin{cases} 1, & x < 2t/3; \\ 0, & x \geq 2t/3. \end{cases}$$

Weak Solutions of Non-viscous Burger's Equation

For smooth u, both the Burger's equation

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} = 0$$

and the equation

$$\frac{\partial u^2}{\partial t} + \frac{2}{3}\frac{\partial u^3}{\partial x} = 0$$

are equivalent to the first order nonlinear hyperbolic equation

$$u_t + uu_x = 0.$$

Key Ingredients of a Conservation Law

The key ingredients of a conservation law are

- the conservative quantities and the corresponding fluxes;
- the integral form equations based on physical conservation laws;
- Entropy condition — to distinguish the physical solution from the others.

Remark: For a nonlinear problem, the weak solution is generally not unique even for a conservation law.

$\mathbb{L}^2$ Is a Better Space to Investigate Hyperbolic Equations

- The upwind scheme of the advection equation satisfies the maximum principle, if CFL condition is satisfied;
- However, for general hyperbolic equations (systems), the solution do not satisfy the maximum principle.
- Hyperbolic equations (systems) are often used to characterize the propagation and evolution of waves.
- Fourier modes are waves of various frequencies propagating at their own characteristic speeds.

It is natural to apply the Fourier method to analyze the $\mathbb{L}^2$ stability and accuracy of finite difference schemes for hyperbolic equations (systems).

Amplification Factors and $\mathbb{L}^2$ stability of the Upwind Scheme (3.2.14)

Substituting the Fourier mode $U_j^m = \lambda_k^m e^{\mathrm{i}kjh}$ into the upwind scheme $U_j^{m+1} = (1-|\nu|)U_j^m + |\nu| U_{j-\mathrm{sign}(a)}^m$ (3.2.14) yields the characteristic equation of the scheme

$$\lambda_k = (1-|\nu|) + |\nu|\, e^{-\mathrm{sign}(a)\,\mathrm{i}kh}.$$

Hence,
$$\begin{aligned} |\lambda_k|^2 &= [(1-|\nu|) + |\nu|\cos kh]^2 + [|\nu|\sin kh]^2 \\ &= 1 - 4|\nu|(1-|\nu|)\sin^2\frac{1}{2}kh. \end{aligned}$$

consequently, for any k, $|\lambda_k| \le 1$ as long as $|\nu| \le 1$. This shows that, for the upwind scheme, (3.2.14) CFL condition is not only a necessary but also a sufficient condition for its $\mathbb{L}^2$ stability.

(Let L be the length of the domain I, then $h = LN^{-1}$, $k = k'\pi L^{-1}$, where the frequency $-N+1 \le k' \le N$.)

Convergence of the Upwind Scheme (3.2.14)

1. CFL condition $|\nu| \leq 1 \Rightarrow \mathbb{L}^2$ stability;
2. more precisely, $\|e^{m+1}\|_2 \leq \|e^0\|_2 + \tau \sum_{l=0}^{m} \|T^l\|_2$;
3. Further more, if $\lim_{\tau \to 0} \int_0^{t_{max}} \|Tu(\cdot,\, t)\|_2 \, dt = 0$, then the upwind scheme is convergent.

In applications, the regularity of the solution u is not always available. When weak solutions is involved, the truncation error above does not make much sense.

An alternative approach: Analytical properties of a difference scheme can often be explored by its errors on the amplitudes and phase angles of Fourier mode solutions.

Amplitude Errors of the Upwind Scheme

If $|\nu| < 1$ is satisfied, $|\lambda_k|^2 = 1 - 4|\nu|(1 - |\nu|)\sin^2 \frac{1}{2}kh < 1$, $\forall k$.

1. $|\lambda_k| = 1 - O(k^2h^2)$ for low frequencies, *i.e.* $kh \ll 1$;
2. $|\lambda_k| = \sqrt{1 - 4|\nu|(1 - |\nu|)}$ for the highest frequency $k = \pi/h$;
3. The higher the frequency, the faster it decays;
4. The numerical solution contains less and less high frequency modes as m increases.
5. For any fixed k, the global approximation error of the upwind scheme on the amplitude is $O(h)$, since the amplitude of the Discrete Fourier mode solution is given by $(1 - O(k^2h^2))^{\tau^{-1}t_{\max}} = 1 - \tau^{-1}t_{\max}O(k^2h^2) = 1 - O(h)$.

Establishment of Lax-Wendroff and Beam-Warming Schemes — 1

Method 1: Characteristic method + 2nd order interpolation.

- The Lagrange quadratic interpolation formula

$$\hat{f}(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2).$$

- The Lax-Wendroff scheme:

$$U_j^{m+1} = -\frac{1}{2}\nu(1-\nu)U_{j+1}^m + (1-\nu^2)U_j^m + \frac{1}{2}\nu(1+\nu)U_{j-1}^m.$$

- The Beam-Warming scheme:

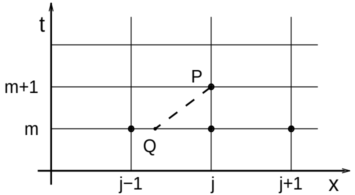
$$U_j^{m+1} = \frac{1}{2}(1-\nu)(2-\nu)U_j^m + \nu(2-\nu)U_{j-1}^m - \frac{1}{2}\nu(1-\nu)U_{j-2}^m.$$

Establishment of Lax-Wendroff and Beam-Warming Schemes — 2

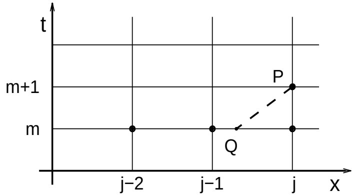
Method 2: Discrete the leading term of the truncation error.

For $a > 0$, the leading term of the truncation error of the upwind scheme is $-\frac{1}{2}ah(1-\nu)u_{xx}$,

- The Lax-Wendroff scheme: substitute $u_{xx}|_j^m$ by $h^{-2}\delta_x^2 u_j^m$.
- The Beam-Warming scheme: substitute $u_{xx}|_j^m$ by $h^{-2}\delta_x^2 u_{j-1}^m$.



Lax−Wendroff



Beam−Warming (a>0)

Establishment of Lax-Wendroff and Beam-Warming Schemes — 3

Method 3: Taylor series expansion with respect to τ + the equation + difference approximations.

1. By the Taylor series expansion
$$u_j^{m+1} = \left[u + \tau u_t + \frac{1}{2}\tau^2 u_{tt}\right]_j^m + O(\tau^3).$$

2. By the advection equation $u_t = -au_x$, $u_{tt} = a^2 u_{xx}$, etc.,
$$u_j^{m+1} = \left[u - a\tau u_x + \frac{1}{2}a^2\tau^2 u_{xx}\right]_j^m + O(\tau^3).$$

Establishment of Lax-Wendroff and Beam-Warming Schemes — 3

3. The Lax-Wendroff scheme: $\partial_x \sim \frac{\triangle_{0x}}{2h}$, $\partial_x^2 \sim \frac{\delta_x^2}{h^2}$.

4. The Beam-Warming scheme: first $\partial_x \sim \frac{\triangle_{-x}}{h}$, yields

$$u_j^{m+1} = u_j^m - a\,\tau \frac{u_j^m - u_{j-1}^m}{h} + \left[\frac{1}{2}(a^2\tau^2 - a\tau h)u_{xx}\right]_j^m + O(\tau^3).$$

then, substitute $u_{xx}|_j^m$ by $h^{-2}\delta_x^2 u_{j-1}^m$.

$\mathbb{L}^2$ Stability of Lax-Wendroff and Beam-Warming Schemes

1. Truncation error of L-W & B-W Schemes: $O(\tau^2 + h^2)$.
2. CFL condition for L-W Scheme: $|\nu| \leq 1$ (see stencil).
3. CFL condition for B-W Scheme: $|\nu| \leq 2$ (see stencil).

$\mathbb{L}^2$ Stability of Lax-Wendroff and Beam-Warming Schemes

4. Characteristic equation for L-W Scheme (see (3.2.41)): $\lambda_k = 1 - \mathrm{i}\nu \sin kh - 2\nu^2 \sin^2 \frac{1}{2}kh$ with its amplitude $|\lambda_k|^2 = 1 - 4\nu^2(1-\nu^2)\sin^4 \frac{1}{2}kh$.

5. Characteristic equation for B-W Scheme ($a > 0$, see (3.2.39)): $\lambda_k = 1 - \nu + \nu e^{-\mathrm{i}kh} - \frac{1}{2}\nu(1-\nu)e^{-\mathrm{i}kh}\left(e^{\mathrm{i}kh} - 2 + e^{-\mathrm{i}kh}\right)$, or $\lambda_k = e^{-\mathrm{i}kh}\left(1 - 2(1-\nu)^2\sin^2\frac{1}{2}kh + \mathrm{i}(1-\nu)\sin kh\right)$ with $|\lambda_k|^2 = 1 - 4\nu(2-\nu)(1-\nu)^2\sin^4\frac{kh}{2}$ $(= 1 - 4(\nu-1)^2(1-(\nu-1)^2)\sin^4\frac{kh}{2})$.

6. If CFL condition is satisfied, both the Lax-Wendroff scheme and the Beam-Warming scheme are $\mathbb{L}^2$ stable.

(Let L be the length of the domain I, then $h = LN^{-1}$, $k = k'\pi L^{-1}$, where the frequency $-N+1 \leq k' \leq N$.)

The Leap-frog Scheme for the Advection Equation

A typical three-time-level scheme for the advection equation is the leap-frog scheme $\dfrac{U_j^{m+1} - U_j^{m-1}}{2\tau} + a\,\dfrac{U_{j+1}^m - U_{j-1}^m}{2h} = 0.$ or equivalently $U_j^{m+1} = U_j^{m-1} - \nu(U_{j+1}^m - U_{j-1}^m)$.

1. CFL condition of the leap-frog scheme: $|\nu| = |a|\tau/h \le 1$.
2. Truncation error: $Tu_j^m = \frac{1}{6}ah^2(1-\nu^2)u_{xxx}|_j^m + O(h^4)$.
3. Characteristic equation: $\lambda_k^2 + 2\mathrm{i}\nu\lambda_k \sin kh - 1 = 0$.
4. Amplification factors: $\lambda_{k\pm} = -\mathrm{i}\nu \sin kh \pm \sqrt{1-\nu^2\sin^2 kh}$.
5. If $|\nu| > 1$, for $kh = \pi/2$, $\max\{|\lambda_{k\pm}|\} = |\nu| + \sqrt{\nu^2 - 1} > 1$.
6. If $|\nu| \le 1$, $|\lambda_{k+}| = |\lambda_{k-}| = 1$, no damping on Fourier modes.
7. The leap-frog scheme is $\mathbb{L}^2$ stable $\Leftrightarrow |\nu| \le 1$.

Inflow, Outflow Boundaries and Numerical Boundary Conditions

For initial-boundary value problems of hyperbolic equations, numerical boundary condition is also an important issue.

1. Inflow boundary: where the characteristic evolves into Ω.
2. Outflow boundary: where the characteristic goes out of Ω.
3. For the advection equation with $a > 0$, the left boundary is inflow, and the right boundary is outflow.
4. On inflow boundary, the boundary condition(s) for the PDE will naturally provide boundary condition(s) for FDM.

Inflow, Outflow Boundaries and Numerical Boundary Conditions

5. Additional numerical boundary conditions are often required on the outflow boundary point by difference schemes.
6. For example, both the Lax-Wendroff scheme and the leap-frog scheme need a numerical boundary condition at the outflow boundary.
7. Use of one sided schemes, such as the upwind scheme and the Beam-warming scheme, on the outflow boundary, can avoid the need for numerical boundary conditions there.

Zero Order Extrapolation and Non-reflection Boundary Conditions

If a numerical boundary condition is required at the outflow right boundary x_N, then, the simplest way we can do is to

- introduce a ghost node x_{N+1};
- apply the zero order extrapolation formula: $U_{N+1}^m = U_N^m$;
- couple the numerical boundary condition with the scheme used on the interior nodes.

The numerical boundary condition so obtained is called a non-reflection boundary condition or an absorbtion boundary condition. Higher order extrapolations are not recommended because of numerical oscillations.

Conservation Law and Its Cell Average Form

1. Differential form of 1D first order scalar conservation law:
$$\frac{\partial u(x,t)}{\partial t} + \frac{\partial f(x,t,u(x,t))}{\partial x} = 0, \qquad x \in I \subset \mathbb{R}, \quad t > 0, \tag{3.3.1}$$
2. Integral form of 1D first order scalar conservation law:
$$\int_{x_l}^{x_r} u(x,t_a)\,dx = \int_{x_l}^{x_r} u(x,t_b)\,dx - \Bigg[\int_{t_b}^{t_a} f(x_r,t,u(x_r,t))\,dt - \int_{t_b}^{t_a} f(x_l,t,u(x_l,t))\,dt\Bigg], \quad \forall x_l < x_r,\ 0 \le t_b < t_a. \tag{3.3.2}$$
3. $\bar{u}_j^m$: the integral average of u on $(x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}})$ at t_m;
4. $\bar{f}_{j-\frac{1}{2}}^{m+\frac{1}{2}}$: the integral average of the flux f on (t_m, t_{m+1}) at $x_{j-\frac{1}{2}}$;
5. $$\bar{u}_j^{m+1} = \bar{u}_j^m - \frac{\tau_m}{h_j}\left[\bar{f}_{j+\frac{1}{2}}^{m+\frac{1}{2}} - \bar{f}_{j-\frac{1}{2}}^{m+\frac{1}{2}}\right], \forall\, j,\ \forall\, m \ge 0. \tag{3.3.3}$$

Numerical Flux and Discrete Conservation Law

1. $U_j^m \sim \bar{u}_j^m$, numerical flux $F_{j\pm\frac{1}{2}}^{m+\frac{1}{2}} \sim \bar{f}_{j\pm\frac{1}{2}}^{m+\frac{1}{2}}$;
2. Numerical flux is of the following (translation invariant) form:

$$F_{j+\frac{1}{2}}^{m+\frac{1}{2}} = F\left([x_{j-p}, U_{j-p}^m, U_{j-p}^{m+1}], \cdots, [x_{j+q}, U_{j+q}^m, U_{j+q}^{m+1}]; t_m, t_{m+1}\right), \tag{3.3.5}$$

3. Conservative difference scheme for the conservation law:

$$U_j^{m+1} = U_j^m - \frac{\tau_m}{h_j}\left[F_{j+\frac{1}{2}}^{m+\frac{1}{2}} - F_{j-\frac{1}{2}}^{m+\frac{1}{2}}\right];$$

4. Discrete conservation law: for all $j_l < j_r$, $k > l \geq 0$,

$$\sum_{j=j_l}^{j_r} h_j\, U_j^k = \sum_{j=j_l}^{j_r} h_j\, U_j^l - \left[\sum_{m=l}^{k-1} \tau_m\, F_{j_r+\frac{1}{2}}^{m+\frac{1}{2}} - \sum_{m=l}^{k-1} \tau_m\, F_{j_l-\frac{1}{2}}^{m+\frac{1}{2}}\right]. \tag{3.3.6}$$

Consistent Conservative Scheme

Definition

A difference scheme is said to be a consistent conservative scheme for the conservation law

$$\int_{x_l}^{x_r} u(x,t_a)\,dx = \int_{x_l}^{x_r} u(x,t_b)\,dx - \left[\int_{t_b}^{t_a} f(x_r,t,u(x_r,t))\,dt - \int_{t_b}^{t_a} f(x_l,t,u(x_l,t))\,dt\right], \ \ \forall x_l < x_r,\ 0 \le t_b < t_a.$$

if its numerical flux is of the form

$$F_{j+\frac{1}{2}}^{m+\frac{1}{2}} = F\left([x_{j-p},U_{j-p}^{m},U_{j-p}^{m+1}],\cdots,[x_{j+q},U_{j+q}^{m},U_{j+q}^{m+1}];t_m,t_{m+1}\right),$$

and

(1) $F([x,u,u],\cdots,[x,u,u];t,t) = f(x,t,u),\ \forall\,(x,t,u);$

(2) F is continuous, and is Lipschitz continuous with respect to the unknown U.

Conservative is Crucial to a Difference Scheme for a Conservation Law

1. Consider the initial value problem of the Burgers equation:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} = 0, \quad u(x,0) = \begin{cases} 1, & x < 0; \\ 0, & x \geq 0; \end{cases}$$

2. The upwind scheme based on the equivalent nonconservative equation $u_t + uu_x = 0$:

$$U_j^{m+1} = \begin{cases} U_j^m - \frac{\tau}{h} U_j^m \left(U_j^m - U_{j-1}^m \right), & U_j^m \geq 0, \\ U_j^m - \frac{\tau}{h} U_j^m \left(U_{j+1}^m - U_j^m \right), & U_j^m < 0. \end{cases}$$

3. Truncation error $O(\tau + h)$, and stable, in fact satisfies the maximum principle, if CFL condition holds.

Conservative is Crucial to a Difference Scheme for a Conservation Law

4. Numerical solution:
$$U_j^m = U_j^0 = \begin{cases} 1, & j < 0; \\ 0, & j \geq 0, \end{cases} \quad \forall m \geq 0.$$

5. Numerical solution converges to $\tilde{u}(x,t) = u(x,0)$, $\forall t > 0$, $\forall x$.

6. $0 = s[\tilde{u}] \neq [f] = 1/2$, Rankine-Hugoniot jump condition is not satisfied, therefore, $\tilde{u}$ is not a weak solution.

Conservative is Crucial to a Difference Scheme for a Conservation Law

For a nonlinear conservation law

7. Non-conservative schemes can converge to a function, which is not a weak solution.
8. Consistency and stability of a finite difference scheme do not necessarily lead to the convergence of the scheme.
9. On the other hand, it can be shown, for a consistent conservative scheme to a conservation law, if the numerical solutions converge in certain specified sense, then the limit function must be a weak solution to the conservation law.

Upwind Numerical Flux and Conservative Upwind Scheme

The finite volume method is a natural way to establish conservative difference schemes. For simplicity, we consider $u_t + f(u)_x = 0$ with f smooth, and uniform grids in tensor product form.

1. Rewrite the conservation law in the form $u_t + f'(u)u_x = 0.$ (3.3.7)
2. The numerical characteristic on the cell interface $x_{j+\frac{1}{2}}$:

$$a^m_{j+\frac{1}{2}} = \begin{cases} \dfrac{f(U^m_{j+1}) - f(U^m_j)}{U^m_{j+1} - U^m_j}, & \text{if } \; U^m_{j+1} \neq U^m_j; \\ 0, & \text{if } \; U^m_{j+1} = U^m_j, \end{cases} \tag{3.3.8}$$

3. CFL condition: $\max_{u\in\mathcal{U}} |f'(u)|\,\tau \leq h$, ($\mathcal{U}$: the solution set). (3.3.9)

Upwind Numerical Flux and Conservative Upwind Scheme

4. Define the numerical flux by (the method of characteristics)

$$F_{j+\frac{1}{2}}^{m+\frac{1}{2}} = \begin{cases} f(U_j^m), & \text{if } \ a_{j+\frac{1}{2}}^m \geq 0; \\ f(U_{j+1}^m), & \text{if } \ a_{j+\frac{1}{2}}^m \leq 0. \end{cases} \tag{3.3.10}$$

This leads to the upwind scheme for the scalar conservation law:

$$\begin{aligned} U_j^{m+1} = U_j^m - \frac{\tau}{2h}\Big\{ &\Big[\Big(1+\mathrm{sign}\Big(a_{j+\frac{1}{2}}^m\Big)\Big) f(U_j^m) + \Big(1-\mathrm{sign}\Big(a_{j+\frac{1}{2}}^m\Big)\Big) f(U_{j+1}^m)\Big] \\ &- \Big[\Big(1+\mathrm{sign}\Big(a_{j-\frac{1}{2}}^m\Big)\Big) f(U_{j-1}^m) + \Big(1-\mathrm{sign}\Big(a_{j-\frac{1}{2}}^m\Big)\Big) f(U_j^m)\Big]\Big\}. \end{aligned} \tag{3.3.11}$$

Upwind Numerical Flux and Conservative Upwind Scheme

5. For $f(u) = au$ ($a =$constant), the scheme is upwind.
6. The scheme is a consistent conservative difference scheme.
7. The integral averages $\bar{u}_j^{m+1}$ and $\bar{u}_j^m$ are calculated by the middle point quadrature rule.
8. The integral averages of the flux $\bar{f}_{j\pm\frac{1}{2}}^{m+\frac{1}{2}}$ are calculated by a one point (generally not the middle point) quadrature rule.
9. Consequently, the truncation error of the upwind scheme is generally only first order.

Middle Point Quadrature and a Conservative Lax-Wendroff Scheme

To achieve higher order accuracy, the middle point quadrature rule can be used to calculate the integral averages of the flux, so that

$$F_{j+\frac{1}{2}}^{m+\frac{1}{2}} = f(U_{j+\frac{1}{2}}^{m+\frac{1}{2}})$$

with a properly defined finite difference operator $R_{j+\frac{1}{2}}$ satisfying

$$U_{j+\frac{1}{2}}^{m+\frac{1}{2}} = R_{j+\frac{1}{2}}(U^m), \quad u_{j+\frac{1}{2}}^{m+\frac{1}{2}} = R_{j+\frac{1}{2}}(u^m) + O(\tau^2 + h^2).$$

1. The two steps Richtmyer scheme:

$$U_{j+\frac{1}{2}}^{m+\frac{1}{2}} = \frac{1}{2}\left(U_j^m + U_{j+1}^m\right) - \frac{\tau}{2h}\left[f(U_{j+1}^m) - f(U_j^m)\right] \triangleq R_{j+\frac{1}{2}}(U^m), \tag{3.3.12}$$

$$U_j^{m+1} = U_j^m - \frac{\tau}{h}\left[f(U_{j+\frac{1}{2}}^{m+\frac{1}{2}}) - f(U_{j-\frac{1}{2}}^{m+\frac{1}{2}})\right]. \tag{3.3.13}$$

Middle Point Quadrature and a Conservative Lax-Wendroff Scheme

2. Let the integral averages $\bar{u}_j^{m+1}$ and $\bar{u}_j^m$ be calculated by the 2nd order accurate middle point quadrature rule.
3. for piecewise constant $\bar{u}_j^m$, $\bar{u}(x_j, t) = \bar{u}_j^m$, $t_m \le t \le t_{m+\frac{1}{2}}$,
4. by Taylor expansion, $u_{j+\frac{1}{2}}^{m+\frac{1}{2}} = u_{j+\frac{1}{2}}^m - \frac{\tau}{2} f(u_{j+\frac{1}{2}}^m)_x + O(\tau^2)$.
5. $f(u_{j+\frac{1}{2}}^m)_x = \frac{f(u_{j+1}^m) - f(u_j^m)}{h} + O(h^2) \Rightarrow u_{j+\frac{1}{2}}^{m+\frac{1}{2}} = R_{j+\frac{1}{2}}(u^m) + O(\tau^2 + h^2)$.
6. Consequently, the truncation error of the scheme is 2nd order.
7. For $f(u) = au$ ($a =$constant), the scheme reduces to the Lax-Wendroff scheme.

Another Consistent Conservative Lax-Wendroff Scheme

There are many ways to extend the Lax-Wendroff scheme to a conservative scheme for the scalar conservation law.

For example, if f is smooth, by $u_t = -f(u)_x$ and $u_{tt} = -(f(u)_t)_x = (f'(u)f(u)_x)_x$, we can establish the Lax-Wendroff scheme for the scalar conservation law

$$U_j^{m+1} = U_j^m - \frac{\tau}{2h}\left[f(U_{j+1}^m) - f(U_{j-1}^m)\right]$$
$$+ \frac{\tau^2}{2h^2}\left[a_{j+\frac{1}{2}}^m \left(f(U_{j+1}^m) - f(U_j^m)\right) - a_{j-\frac{1}{2}}^m \left(f(U_j^m) - f(U_{j-1}^m)\right)\right], \quad (3.3.14)$$

where $a_{j\pm\frac{1}{2}}^m = f'(\frac{1}{2}(U_j^m + U_{j\pm 1}^m))$. It can be shown :

Another Consistent Conservative Lax-Wendroff Scheme

1. For $f(u) = au$ (constant a), the scheme reduces to the Lax-Wendroff scheme.
2. It is a consistent conservative scheme for the scalar conservation law.
3. The truncation error of the scheme is 2nd order.

Numerical Initial-Boundary Conditions To Maintain Global Conservation

1. An initial-boundary value problem of a conservation law ($f' > 0$):

$$\frac{\partial u}{\partial t}+\frac{\partial f(u)}{\partial x}=0,\quad x\in(0,1),\ t>0,\begin{cases} u(x,0)=u^0(x), & x\in(0,1);\\ u(0,t)=u_0(t), & t\geq 0.\end{cases} \tag{3.3.15-17}$$

2. Global conservation (or balance) of weak solutions:

$$\int_0^1 u(x,t)\,dx=\int_0^1 u(x,0)\,dx-\left[\int_0^t f(u(1,t))\,dt-\int_0^t f(u(0,t))\,dt\right]. \tag{3.3.18}$$

3. Conservative scheme: $U_j^{m+1}=U_j^m-\tau\, h^{-1}\left[F^m_{j+\frac{1}{2}}-F^m_{j-\frac{1}{2}}\right].$

Numerical Initial-Boundary Conditions To Maintain Global Conservation

4. Numerical initial and boundary conditions ($f' > 0$):

$$U_j^0 = \frac{1}{h}\int_{j-\frac{1}{2}}^{j+\frac{1}{2}} u^0(x)\,dx,\ \forall\, 1 \le j \le N;\ F_{\frac{1}{2}}^m = \frac{1}{\tau}\int_{t_m}^{t_{m+1}} f(u_0(t))\,dt,\ \forall\, m \ge 0. \tag{3.3.19}$$

5. We can also define the value of U_0^m on the ghost node x_0 by making use of the relation

$$F(U_0^m, U_1^m) = F_{\frac{1}{2}}^m = \frac{1}{\tau}\int_{t_m}^{t_{m+1}} f(u_0(t))\,dt. \tag{3.3.21}$$

Numerical Initial-Boundary Conditions To Maintain Global Conservation

6. Global property of the numerical solution:

$$h\sum_{j=1}^{N} U_j^0 = \int_0^1 u^0(x)\,dx, \quad \tau \sum_{l=m}^{m+k} F_{\frac{1}{2}}^l = \int_{t_m}^{t_{m+k}} f(u_0(t))\,dt.$$

7. Numerical boundary condition on outflow boundary — construction of upwind numerical flux:

   (1) 1st-order upwind numerical flux $F_{N+\frac{1}{2}}^m = f(U_N^m)$,

   (may also be viewed as resulted from the zero order extrapolation $U_{N+1}^m = U_N^m$);

   (2) 2nd-order upwind numerical flux $\hat{F}_{N+\frac{1}{2}}^m = \hat{F}(U_{N-1}^m, U_N^m)$.

   (say the Beam-Warming Flux; may also be viewed as resulted from the 1st-order extrapolation.)

Advection-Diffusion Equation — A Model Problem

- An initial value problem of a 1D constant-coefficient advection-diffusion equation ($a > 0,\ c > 0$): $u_t + au_x = cu_{xx}$, (3.4.1)
  $x \in \mathbb{R},\ t > 0; \quad u(x, 0) = u^0(x),\ x \in \mathbb{R}$.
- By a change of variables $y = x - at$ and $v(y, t) \triangleq u(y + at, t)$,
  $v_t = cv_{yy}, \quad y \in \mathbb{R},\ \ t > 0;\ v(x, 0) = u^0(x),\ x \in \mathbb{R}$.

Characteristic global properties of the solution u:

1. There is a characteristic speed as in the advection equation, which plays an important role to the solution, especially when $|a| \gg c$ (advection dominant).
2. Along the characteristic, the solution behaves like a parabolic solution (dissipation and smoothing).

Central Explicit Scheme for the Advection-Diffusion Equation

The simplest difference scheme is the central explicit scheme

$$\frac{U_j^{m+1} - U_j^m}{\tau} + a\frac{U_{j+1}^m - U_{j-1}^m}{2h} = c\frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{h^2}.$$

or equivalently ($\nu = a\tau/h$, $\mu = c\tau/h^2$)

$$U_j^{m+1} = (\mu - \frac{1}{2}\nu)U_{j+1}^m + (1-2\mu)U_j^m + (\mu + \frac{1}{2}\nu)U_{j-1}^m,$$

1. Condition for the maximum principle: $\mu = \frac{c\tau}{h^2} \leq \frac{1}{2}$, $h \leq \frac{2c}{a}$.
2. Amplification factor $\lambda_k = 1 - 2\mu(1 - \cos kh) - \mathrm{i}\nu \sin kh$.

Note: L^2-stable $\Leftrightarrow |\lambda_k| \leq 1 + O(\tau)$; L^2-strongly stable $\Leftrightarrow |\lambda_k| \leq 1$.

$\mathbb{L}^2$ Strong Stability Conditions of the Central Explicit Scheme

3. $|\lambda_k|^2 = 1 - (1 - \cos kh)[4\mu - 4\mu^2(1 - \cos kh) - \nu^2(1 + \cos kh)]$.

4. Strongly stable $\Leftrightarrow 4\mu - 4\mu^2(1 - \cos kh) - \nu^2(1 + \cos kh) \geq 0$
$\Leftrightarrow 4\mu - 2\nu^2 + (\nu^2 - 4\mu^2)(1 - \cos kh) \geq 0$, if $\cos kh \neq 1$.

5. $\mathbb{L}^2$ strongly stable $\Leftrightarrow 4\mu - 2\nu^2 + (\nu^2 - 4\mu^2)(1 - \cos kh) \geq 0$.

6. Take $k = \pi h^{-1}$, we are lead to a necessary condition:
$4\mu(1 - 2\mu) = 4\mu - 2\nu^2 + 2(\nu^2 - 4\mu^2) \geq 0 \Leftrightarrow \mu \leq \frac{1}{2}$.

$\mathbb{L}^2$ Strong Stability Conditions of the Central Explicit Scheme

7. Let $kh \to 0 \Rightarrow \nu^2 \le 2\mu \Leftrightarrow \tau \le \frac{2c}{a^2}$, a necessary condition.

8. $g(\xi) = 4\mu - 2\nu^2 + (\nu^2 - 4\mu^2)\xi$ is monotonic with respect to ξ, and $g(0) \ge 0$, $g(2) \ge 0$, if the two necessary conditions hold.

9. $g(\xi) \ge 0$, $\forall \xi \in [0, 2]$, if the two necessary conditions hold.

10. $\mathbb{L}^2$ strongly stable $\Leftrightarrow$ $\mu \le \frac{1}{2}$ and $\tau \le \frac{2c}{a^2}$.

Péclet Number and the Stability of the Central Explicit Scheme

Under the condition $\mu \leq 1/2$,

11. the maximum principal holds if and only if the Péclet number $\frac{|a|h}{c}$ satisfies $\frac{|a|h}{c} \leq 2$.

12. $\frac{|a|h}{c} \leq 2 \Rightarrow \tau \leq h^2/(2c) \leq 4c^2/(2ca^2) = 2c/a^2 \Rightarrow$ the second condition for $\mathbb{L}^2$ strong stability holds automatically.

13. When $\frac{|a|h}{c} > 2$, the scheme does not satisfy the maximum principle, and the condition $\tau \leq \frac{2c}{a^2}$ substantially restricts the time step size for the scheme to be $\mathbb{L}^2$ strongly stable .

Note: The truncation error of the Central Explicit Scheme is $O(\tau + h^2)$.

Modified Central Explicit Scheme for the Advection-Diffusion Equation

By $u_t = -au_x + cu_{xx}$, $u_{tt} = a^2 u_{xx} - 2ac\partial_x^3 u + c^2 \partial_x^4 u$, for $c \ll |a|$:

$$\begin{aligned} u_j^{m+1} &= \left[u + \tau u_t + \frac{\tau^2}{2} u_{tt} \right]_j^m + O(\tau^3) \\ &= \left[u - a\tau u_x + \left(c + \frac{a^2\tau}{2} \right) \tau u_{xx} \right]_j^m + O(c\tau^2 + \tau^3). \end{aligned}$$

This leads to the modified central explicit scheme

$$\frac{U_j^{m+1} - U_j^m}{\tau} + a\frac{U_{j+1}^m - U_{j-1}^m}{2h} = \left(c + \frac{1}{2} a^2 \tau \right) \frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{h^2}.$$

Modified Central Explicit Scheme for the Advection-Diffusion Equation

1. Local truncation error $O(c\tau + \tau^2 + h^2)$.

2. By the results of the central explicit scheme (with $\tilde{c} = c + \frac{1}{2}a^2\tau$):

$$\begin{cases} \left(c + \dfrac{1}{2}a^2\tau\right)\dfrac{\tau}{h^2} \le \dfrac{1}{2}, \quad h \le \dfrac{2c + a^2\tau}{|a|}, & \Leftrightarrow \text{maximum principle;} \\ \left(c + \dfrac{1}{2}a^2\tau\right)\dfrac{\tau}{h^2} \le \dfrac{1}{2}, \quad \tau \le \dfrac{2c + a^2 h}{a^2}, & \Leftrightarrow \mathbb{L}^2 \text{ strong stability.} \end{cases}$$

3. Compared with the central explicit scheme, to satisfy the maximum principle, the condition on the spatial grid size is relaxed, but the condition on the grid ratio is more restricted.

Stability Conditions for the Modified Central Explicit Scheme

4. The second condition for the $\mathbb{L}^2$ strong stability holds automatically.

5. The necessary and sufficient stability conditions:

$$\begin{cases} \left(c + \frac{1}{2}a^2\tau\right)\frac{\tau}{h^2} \leq \frac{1}{2}, \quad h \leq \frac{2c + a^2\tau}{a}, & \Leftrightarrow \text{maximum principle;} \\ \left(c + \frac{1}{2}a^2\tau\right)\frac{\tau}{h^2} \leq \frac{1}{2}, & \Leftrightarrow \mathbb{L}^2 \text{ strong stability.} \end{cases}$$

The Upwind Scheme for the Advection-Diffusion Equation

The upwind scheme ($a > 0$)

$$\frac{U_j^{m+1} - U_j^m}{\tau} + a\frac{U_j^m - U_{j-1}^m}{h} = c\frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{h^2},$$

or equivalently (compare the central explicit scheme)

$$\frac{U_j^{m+1} - U_j^m}{\tau} + a\frac{U_{j+1}^m - U_{j-1}^m}{2h} = \left(c + \frac{1}{2}ah\right)\frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{h^2}.$$

The Upwind Scheme for the Advection-Diffusion Equation

1. Local truncation error $O(\tau + h)$.
2. By the results of the central explicit scheme (with $\tilde{c} = c + \frac{1}{2}ah$):

$$\begin{cases} (2c+ah)\dfrac{\tau}{h^2} \leq 1, \quad h \leq \dfrac{2c+ah}{a}, & \Leftrightarrow \text{maximum principle;} \\ (2c+ah)\dfrac{\tau}{h^2} \leq 1, \quad \tau \leq \dfrac{2c+ah}{a^2}, & \Leftrightarrow \mathbb{L}^2 \text{ strong stability.} \end{cases}$$

3. The second condition for the maximum principle holds automatically.

The Stability Condition for the Upwind Scheme

4. The second condition for the $\mathbb{L}^2$ strong stability is a consequence of the first, since

$$\frac{a^2\tau}{2c+ah} = \frac{a^2h^2}{2c+ah}\,\frac{\tau}{h^2} \leq \frac{(2c+ah)^2}{2c+ah}\,\frac{\tau}{h^2} = (2c+ah)\frac{\tau}{h^2},$$

5. The necessary and sufficient stability condition:

$$\begin{cases} (2c+ah)\dfrac{\tau}{h^2} \leq 1, & \Leftrightarrow \text{maximum principle;} \\ (2c+ah)\dfrac{\tau}{h^2} \leq 1, & \Leftrightarrow \mathbb{L}^2 \text{ strong stability.} \end{cases}$$

A Crank-Nicolson Type Scheme for the Advection-Diffusion Equation

The Crank-Nicolson scheme for the advection-diffusion equation:

$$\frac{U_j^{m+1} - U_j^m}{\tau} + \frac{a}{2}\left[\frac{U_{j+1}^m - U_{j-1}^m}{2h} + \frac{U_{j+1}^{m+1} - U_{j-1}^{m+1}}{2h}\right]$$
$$= \frac{c}{2}\left[\frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{h^2} + \frac{U_{j+1}^{m+1} - 2U_j^{m+1} + U_{j-1}^{m+1}}{h^2}\right].$$

or

$$(1+\mu)U_j^{m+1} = (1-\mu)U_j^{m+1} + \frac{1}{2}(\mu - \frac{\nu}{2})\left(U_{j+1}^m + U_{j+1}^{m+1}\right)$$
$$+ \frac{1}{2}(\mu + \frac{\nu}{2})\left(U_{j-1}^m + U_{j-1}^{m+1}\right).$$

A Crank-Nicolson Type Scheme for the Advection-Diffusion Equation

1. Amplification factor: $\lambda_k = \dfrac{1-\mu(1-\cos kh) - \mathrm{i}\frac{1}{2}\nu\sin kh}{1+\mu(1-\cos kh) + \mathrm{i}\frac{1}{2}\nu\sin kh}$.

2. $|\lambda_k| \le 1$, for all k, unconditionally $\mathbb{L}^2$ strongly stable.

3. Condition for the maximum principle $\mu \le 1$, $h \le \dfrac{2c}{|a|}$, *i.e.* the condition on μ is the same as for the diffusion equation, provided h is less than the Péclet number $\frac{|a|h}{c} \le 2$.

佩克莱数(Peclet number,简称Pe数)是一个在连续传输现象研究中经常用到的一个无量纲数。其物理意义为对流速率与扩散速率之比，其中扩散速率是指在一定浓度梯度驱使下的扩散速率。在物质质量传递的情况下下，Péclet 数是雷诺数（Reynolds number）和施密特数（Schmidt number）的乘积。在热流体传热中，热Peclet数相当于雷诺数（Reynolds number）和普朗特数（Prandtl number）的乘积。